Calculations

1. Fission of one U-235 atom releases about 200 MeV of energy. How many kWh thermal energy is released from fission of 1g ^{235}U?

2. The average uranium concentration in Earth's crust is 2.8 ppm. In typical mines the concentration ranges from 0.02% to 0.4% (determined as U_3O_8), although at certain locations it can be up to 20%. If mined from a deposit with 0.4% uranium (U_3O_8). How much ore must ideally be process to charge an electric vehicle battery of 75 kWh (assume that only U-235 is utilized, electricity is produced with a carnot efficiency of 40%, and that there are no losses in the entire process)

3. According to IAEA, a liter of sea water contains about 33 mg of deuterium. How much fusion energy does that correspond to? (the energy of fusion is 17.6 MeV).

4. In a BWR or PWR, steam is generated with a temperature of about 290 ^oC. (a) If river water used to receive excess heat has a temperature of 20 ^oC, what is the maximum possible conversion efficiency of the reactor's thermal energy into electricity? (b) Nuclear power plants typically have conversion efficiencies of 34 %. Why is this efficiency less than the ideal efficiency?

5. A 1000 MWe nuclear power plant has a thermal conversion efficiency of 33%. (a) How much thermal power is rejected through the condenser to the cooling water? (b) What is the flow rate (kg/s) of the condenser cooling water if the temperature rise of this water is 12 ^oC? Specific heat of water is 4184 J $kg^{-1}K^{-1}$.

6. The energy of U-235 fusion is about 200 MeV. What is the energy of combustion of a methane molecule in the unit eV? A heat of combustion table can be found on Learn.

7. Calculate the mass defect of $^{232}_{90}\mathrm{Th}$. The atomic mass is 232.038 u. Find additional information needed in the list of formulas.